



Chapter 6: Orthogonality and Least squares

Question 1:

1. Since $\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$, $\mathbf{u} \cdot \mathbf{u} = (-1)^2 + 2^2 = 5$, $\mathbf{v} \cdot \mathbf{u} = 4(-1) + 6(2) = 8$, and $\frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} = \frac{8}{5}$.

2. Since $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$, $\mathbf{w} \cdot \mathbf{w} = 3^2 + (-1)^2 + (-5)^2 = 35$, $\mathbf{x} \cdot \mathbf{w} = 6(3) + (-2)(-1) + 3(-5) = 5$, and $\frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}} = \frac{5}{35} = \frac{1}{7}$.

3. Since $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$, $\mathbf{w} \cdot \mathbf{w} = 3^2 + (-1)^2 + (-5)^2 = 35$, and $\frac{1}{\mathbf{w} \cdot \mathbf{w}} \mathbf{w} = \begin{bmatrix} 3/35 \\ -1/35 \\ -1/7 \end{bmatrix}$.

4. Since $\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\mathbf{u} \cdot \mathbf{u} = (-1)^2 + 2^2 = 5$ and $\frac{1}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix}$.

5. Since $\mathbf{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$, $\mathbf{u} \cdot \mathbf{v} = (-1)(4) + 2(6) = 8$, $\mathbf{v} \cdot \mathbf{v} = 4^2 + 6^2 = 52$, and $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \right) \mathbf{v} = \frac{2}{13} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 8/13 \\ 12/13 \end{bmatrix}$.

6. Since $\mathbf{x} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$, $\mathbf{x} \cdot \mathbf{w} = 6(3) + (-2)(-1) + 3(-5) = 5$, $\mathbf{x} \cdot \mathbf{x} = 6^2 + (-2)^2 + 3^2 = 49$, and $\left(\frac{\mathbf{x} \cdot \mathbf{w}}{\mathbf{x} \cdot \mathbf{x}} \right) \mathbf{x} = \frac{5}{49} \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 30/49 \\ -10/49 \\ 15/49 \end{bmatrix}$.

7. Since $\mathbf{w} = \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix}$, $\|\mathbf{w}\| = \sqrt{\mathbf{w} \cdot \mathbf{w}} = \sqrt{3^2 + (-1)^2 + (-5)^2} = \sqrt{35}$.

8. Since $\mathbf{x} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$, $\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{6^2 + (-2)^2 + 3^2} = \sqrt{49} = 7$.

9. A unit vector in the direction of the given vector is

$$\frac{1}{\sqrt{(-30)^2 + 40^2}} \begin{bmatrix} -30 \\ 40 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} -30 \\ 40 \end{bmatrix} = \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}$$

10. A unit vector in the direction of the given vector is

$$\frac{1}{\sqrt{(-6)^2 + 4^2 + (-3)^2}} \begin{bmatrix} -6 \\ 4 \\ -3 \end{bmatrix} = \frac{1}{\sqrt{61}} \begin{bmatrix} -6 \\ 4 \\ -3 \end{bmatrix} = \begin{bmatrix} -6/\sqrt{61} \\ 4/\sqrt{61} \\ -3/\sqrt{61} \end{bmatrix}$$

11. A unit vector in the direction of the given vector is

$$\frac{1}{\sqrt{(7/4)^2 + (1/2)^2 + 1^2}} \begin{bmatrix} 7/4 \\ 1/2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{69/16}} \begin{bmatrix} 7/4 \\ 1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7/\sqrt{69} \\ 2/\sqrt{69} \\ 4/\sqrt{69} \end{bmatrix}$$

12. A unit vector in the direction of the given vector is

$$\frac{1}{\sqrt{(8/3)^2 + 2^2}} \begin{bmatrix} 8/3 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{100/9}} \begin{bmatrix} 8/3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}$$

13. Since $\mathbf{x} = \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$, $\|\mathbf{x} - \mathbf{y}\|^2 = [10 - (-1)]^2 + [-3 - (-5)]^2 = 125$ and $\text{dist}(\mathbf{x}, \mathbf{y}) = \sqrt{125} = 5\sqrt{5}$.

14. Since $\mathbf{u} = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix}$, $\|\mathbf{u} - \mathbf{z}\|^2 = [0 - (-4)]^2 + [-5 - (-1)]^2 + [2 - 8]^2 = 68$ and $\text{dist}(\mathbf{u}, \mathbf{z}) = \sqrt{68} = 2\sqrt{17}$.

15. Since $\mathbf{a} \cdot \mathbf{b} = 8(-2) + (-5)(-3) = -1 \neq 0$, $\mathbf{a}$ and $\mathbf{b}$ are not orthogonal.

16. Since $\mathbf{u} \cdot \mathbf{v} = 12(2) + (3)(-3) + (-5)(3) = 0$, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

17. Since $\mathbf{u} \cdot \mathbf{v} = 3(-4) + 2(1) + (-5)(-2) + 0(6) = 0$, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

18. Since $\mathbf{y} \cdot \mathbf{z} = (-3)(1) + 7(-8) + 4(15) + 0(-7) = 1 \neq 0$, $\mathbf{y}$ and $\mathbf{z}$ are not orthogonal.

Question 2:

1. Since $\begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -4 \\ -7 \end{bmatrix} = 2 \neq 0$, the set is not orthogonal.

2. Since $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} = 0$, the set is orthogonal.

3. Since $\begin{bmatrix} -6 \\ -3 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = -30 \neq 0$, the set is not orthogonal.

4. Since $\begin{bmatrix} 2 \\ -5 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix} = 0$, the set is orthogonal.

5. Since $\begin{bmatrix} 3 \\ -2 \\ 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 8 \\ 7 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 8 \\ 7 \\ 0 \end{bmatrix} = 0$, the set is orthogonal.

6. Since $\begin{bmatrix} -4 \\ 1 \\ -3 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 5 \\ -1 \end{bmatrix} = -32 \neq 0$, the set is not orthogonal.

In the following exercises, you can also find the coefficients by solving a linear system

7. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = 12 - 12 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. Since the vectors are non-zero, $\mathbf{u}_1$ and $\mathbf{u}_2$ are linearly independent by Theorem 4. Two such vectors in $\mathbb{R}^2$ automatically form a basis for $\mathbb{R}^2$. So $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $\mathbb{R}^2$. By Theorem 5,

$$\mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{x} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = 3\mathbf{u}_1 + \frac{1}{2}\mathbf{u}_2$$

8. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = -6 + 6 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. Since the vectors are non-zero, $\mathbf{u}_1$ and $\mathbf{u}_2$ are linearly independent by Theorem 4. Two such vectors in $\mathbb{R}^2$ automatically form a basis for $\mathbb{R}^2$. So $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $\mathbb{R}^2$. By Theorem 5,

$$\mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{x} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = -\frac{3}{2} \mathbf{u}_1 + \frac{3}{4} \mathbf{u}_2$$

9. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1 \cdot \mathbf{u}_3 = \mathbf{u}_2 \cdot \mathbf{u}_3 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. Since the vectors are non-zero, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ are linearly independent by Theorem 4. Three such vectors in $\mathbb{R}^3$ automatically form a basis for $\mathbb{R}^3$. So $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. By Theorem 5,

$$\mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{x} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{x} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3 = \frac{5}{2} \mathbf{u}_1 - \frac{3}{2} \mathbf{u}_2 + 2 \mathbf{u}_3$$

10. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1 \cdot \mathbf{u}_3 = \mathbf{u}_2 \cdot \mathbf{u}_3 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. Since the vectors are non-zero, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ are linearly independent by Theorem 4. Three such vectors in $\mathbb{R}^3$ automatically form a basis for $\mathbb{R}^3$. So $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. By Theorem 5,

$$\mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{x} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{x} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3 = \frac{4}{3} \mathbf{u}_1 + \frac{1}{3} \mathbf{u}_2 + \frac{1}{3} \mathbf{u}_3$$

11. Let $\mathbf{y} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$. The orthogonal projection of $\mathbf{y}$ onto the line through $\mathbf{u}$ and the origin is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$, and this vector is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{1}{2} \mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

12. Let $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. The orthogonal projection of $\mathbf{y}$ onto the line through $\mathbf{u}$ and the origin is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$, and this vector is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = -\frac{2}{5} \mathbf{u} = \begin{bmatrix} 2/5 \\ -6/5 \end{bmatrix}$$

13. The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = -\frac{13}{65} \mathbf{u} = \begin{bmatrix} -4/5 \\ 7/5 \end{bmatrix}$$

The component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 14/5 \\ 8/5 \end{bmatrix}$$

$$\text{Thus } \mathbf{y} = \hat{\mathbf{y}} + (\mathbf{y} - \hat{\mathbf{y}}) = \begin{bmatrix} -4/5 \\ 7/5 \end{bmatrix} + \begin{bmatrix} 14/5 \\ 8/5 \end{bmatrix}.$$

14. The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{2}{5} \mathbf{u} = \begin{bmatrix} 14/5 \\ 2/5 \end{bmatrix}$$

The component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -4/5 \\ 28/5 \end{bmatrix}$$

$$\text{Thus } \mathbf{y} = \hat{\mathbf{y}} + (\mathbf{y} - \hat{\mathbf{y}}) = \begin{bmatrix} 14/5 \\ 2/5 \end{bmatrix} + \begin{bmatrix} -4/5 \\ 28/5 \end{bmatrix}.$$

15. The distance from $\mathbf{y}$ to the line through $\mathbf{u}$ and the origin is $\|\mathbf{y} - \hat{\mathbf{y}}\|$. One computes that

$$\mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \frac{3}{10} \begin{bmatrix} 8 \\ 6 \end{bmatrix} = \begin{bmatrix} 3/5 \\ -4/5 \end{bmatrix}$$

so $\|\mathbf{y} - \hat{\mathbf{y}}\| = \sqrt{9/25 + 16/25} = 1$ is the desired distance.

16. The distance from $\mathbf{y}$ to the line through $\mathbf{u}$ and the origin is $\|\mathbf{y} - \hat{\mathbf{y}}\|$. One computes that

$$\mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \begin{bmatrix} -3 \\ 9 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$$

so $\|\mathbf{y} - \hat{\mathbf{y}}\| = \sqrt{36 + 9} = 3\sqrt{5}$ is the desired distance.

17. Let $\mathbf{u} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = 0$, $\{\mathbf{u}, \mathbf{v}\}$ is an orthogonal set. However, $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 1/3$ and

$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 1/2$, so $\{\mathbf{u}, \mathbf{v}\}$ is not an orthonormal set. The vectors $\mathbf{u}$ and $\mathbf{v}$ may be normalized to form the orthonormal set

$$\left\{ \frac{\mathbf{u}}{\|\mathbf{u}\|}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\} = \left\{ \begin{bmatrix} \sqrt{3}/3 \\ \sqrt{3}/3 \\ \sqrt{3}/3 \end{bmatrix}, \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix} \right\}$$

18. Let $\mathbf{u} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = -1 \neq 0$, $\{\mathbf{u}, \mathbf{v}\}$ is not an orthogonal set.

19. Let $\mathbf{u} = \begin{bmatrix} -.6 \\ .8 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} .8 \\ .6 \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = 0$, $\{\mathbf{u}, \mathbf{v}\}$ is an orthogonal set. Also, $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 1$ and

$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 1$, so $\{\mathbf{u}, \mathbf{v}\}$ is an orthonormal set.

20. Let $\mathbf{u} = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = 0$, $\{\mathbf{u}, \mathbf{v}\}$ is an orthogonal set. However, $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 1$ and

$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 5/9$, so $\{\mathbf{u}, \mathbf{v}\}$ is not an orthonormal set. The vectors $\mathbf{u}$ and $\mathbf{v}$ may be normalized to form the orthonormal set

$$\left\{ \frac{\mathbf{u}}{\|\mathbf{u}\|}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\} = \left\{ \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix} \right\}$$

21. Let $\mathbf{u} = \begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{20} \\ 3/\sqrt{20} \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3/\sqrt{10} \\ -1/\sqrt{20} \\ -1/\sqrt{20} \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = 0$, $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an

orthogonal set. Also, $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 1$, $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 1$, and $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} = 1$, so $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an orthonormal set.

22. Let $\mathbf{u} = \begin{bmatrix} 1/\sqrt{18} \\ 4/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} -2/3 \\ 1/3 \\ -2/3 \end{bmatrix}$. Since $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = 0$, $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an

orthogonal set. Also, $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 1$, $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 1$, and $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} = 1$, so $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an orthonormal set.

Question 3:[Orthogonal Projection]

1. The vector in $\text{Span}\{\mathbf{u}_4\}$ is

$$\frac{\mathbf{x} \cdot \mathbf{u}_4}{\mathbf{u}_4 \cdot \mathbf{u}_4} \mathbf{u}_4 = \frac{72}{36} \mathbf{u}_4 = 2\mathbf{u}_4 = \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix}$$

Since $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + \frac{\mathbf{x} \cdot \mathbf{u}_4}{\mathbf{u}_4 \cdot \mathbf{u}_4} \mathbf{u}_4$, the vector

$$\mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{u}_4}{\mathbf{u}_4 \cdot \mathbf{u}_4} \mathbf{u}_4 = \begin{bmatrix} 10 \\ -8 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 4 \\ -2 \end{bmatrix}$$

is in $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

2. The vector in $\text{Span}\{\mathbf{u}_1\}$ is

$$\frac{\mathbf{v} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 = \frac{14}{7} \mathbf{u}_1 = 2\mathbf{u}_1 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix}$$

Since $\mathbf{x} = \frac{\mathbf{v} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3 + c_4 \mathbf{u}_4$, the vector

$$\mathbf{v} - \frac{\mathbf{v} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 = \begin{bmatrix} 4 \\ 5 \\ -3 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -5 \\ 1 \end{bmatrix}$$

is in $\text{Span}\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.

3. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = -1 + 1 + 0 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. The orthogonal projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = \frac{3}{2} \mathbf{u}_1 + \frac{5}{2} \mathbf{u}_2 = \frac{3}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

4. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = -12 + 12 + 0 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. The orthogonal projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = \frac{30}{25} \mathbf{u}_1 - \frac{15}{25} \mathbf{u}_2 = \frac{6}{5} \begin{bmatrix} 3 \\ 4 \\ 0 \\ 0 \end{bmatrix} - \frac{3}{5} \begin{bmatrix} -4 \\ 3 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \\ 0 \end{bmatrix}$$

5. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = 3 + 1 - 4 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. The orthogonal projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = \frac{7}{14} \mathbf{u}_1 - \frac{15}{6} \mathbf{u}_2 = \frac{1}{2} \begin{bmatrix} 3 \\ -1 \\ 2 \\ 2 \end{bmatrix} - \frac{5}{2} \begin{bmatrix} 1 \\ -1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \\ 6 \end{bmatrix}$$

6. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0 - 1 + 1 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. The orthogonal projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = -\frac{27}{18} \mathbf{u}_1 + \frac{5}{2} \mathbf{u}_2 = -\frac{3}{2} \begin{bmatrix} -4 \\ -1 \\ 1 \\ 1 \end{bmatrix} + \frac{5}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \\ 1 \end{bmatrix}$$

7. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = 5 + 3 - 8 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. By the Orthogonal Decomposition Theorem,

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = 0\mathbf{u}_1 + \frac{2}{3}\mathbf{u}_2 = \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix}, \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -7/3 \\ 7/3 \\ 7/3 \end{bmatrix}$$

and $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$, where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

8. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = -1 + 3 - 2 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set. By the Orthogonal Decomposition Theorem,

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = 2\mathbf{u}_1 + \frac{1}{2}\mathbf{u}_2 = \begin{bmatrix} 3/2 \\ 7/2 \\ 1 \end{bmatrix}, \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -5/2 \\ 1/2 \\ 2 \end{bmatrix}$$

and $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$, where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

9. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1 \cdot \mathbf{u}_3 = \mathbf{u}_2 \cdot \mathbf{u}_3 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. By the Orthogonal Decomposition Theorem,

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3 = 2\mathbf{u}_1 + \frac{2}{3}\mathbf{u}_2 - \frac{2}{3}\mathbf{u}_3 = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ -1 \end{bmatrix}$$

and $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$, where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

10. Since $\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1 \cdot \mathbf{u}_3 = \mathbf{u}_2 \cdot \mathbf{u}_3 = 0$, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. By the Orthogonal Decomposition Theorem,

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \frac{\mathbf{y} \cdot \mathbf{u}_3}{\mathbf{u}_3 \cdot \mathbf{u}_3} \mathbf{u}_3 = \frac{1}{3}\mathbf{u}_1 + \frac{14}{3}\mathbf{u}_2 - \frac{5}{3}\mathbf{u}_3 = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}, \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

and $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$, where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

11. Note that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. The Best Approximation Theorem says that $\hat{\mathbf{y}}$, which is the orthogonal projection of $\mathbf{y}$ onto $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, is the closest point to $\mathbf{y}$ in W . This vector is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{y} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \frac{1}{2}\mathbf{v}_1 + \frac{3}{2}\mathbf{v}_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

12. Note that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. The Best Approximation Theorem says that $\hat{\mathbf{y}}$, which is the orthogonal projection of $\mathbf{y}$ onto $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, is the closest point to $\mathbf{y}$ in W . This vector is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{y} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = 3\mathbf{v}_1 + 1\mathbf{v}_2 = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}$$

13. Note that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. By the Best Approximation Theorem, the closest point in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\mathbf{z}$ is

$$\hat{\mathbf{z}} = \frac{\mathbf{z} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{z} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \frac{2}{3} \mathbf{v}_1 - \frac{7}{3} \mathbf{v}_2 = \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}$$

14. Note that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. By the Best Approximation Theorem, the closest point in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\mathbf{z}$ is

$$\hat{\mathbf{z}} = \frac{\mathbf{z} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 + \frac{\mathbf{z} \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \frac{1}{2} \mathbf{v}_1 + 0 \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1/2 \\ -3/2 \end{bmatrix}$$

15. The distance from the point $\mathbf{y}$ in $\mathbb{R}^3$ to a subspace W is defined as the distance from $\mathbf{y}$ to the closest point in W . Since the closest point in W to $\mathbf{y}$ is $\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$, the desired distance is $\|\mathbf{y} - \hat{\mathbf{y}}\|$. One computes that

$$\hat{\mathbf{y}} = \begin{bmatrix} 3 \\ -9 \\ -1 \end{bmatrix}, \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}, \text{ and } \|\mathbf{y} - \hat{\mathbf{y}}\| = \sqrt{40} = 2\sqrt{10}.$$

16. The distance from the point $\mathbf{y}$ in $\mathbb{R}^4$ to a subspace W is defined as the distance from $\mathbf{y}$ to the closest point in W . Since the closest point in W to $\mathbf{y}$ is $\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$, the desired distance is $\|\mathbf{y} - \hat{\mathbf{y}}\|$. One computes that

$$\hat{\mathbf{y}} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}, \text{ and } \|\mathbf{y} - \hat{\mathbf{y}}\| = 8.$$

Question 4: [Gram Schmidt]

1. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - 3\mathbf{v}_1 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix} \right\}.$$

2. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - \frac{1}{2} \mathbf{v}_1 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix} \right\}.$$

3. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - \frac{1}{2} \mathbf{v}_1 = \begin{bmatrix} 3 \\ 3/2 \\ 3/2 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 3/2 \\ 3/2 \end{bmatrix} \right\}.$$

4. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - (-2)\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix} \right\}.$$

5. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - 2\mathbf{v}_1 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix} \right\}.$$

6. Set $\mathbf{v}_1 = \mathbf{x}_1$ and compute that $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - (-3)\mathbf{v}_1 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}$. Thus an orthogonal basis for W is

$$\left\{ \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix} \right\}.$$

Question 5:

9. Call the columns of the matrix $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ and perform the Gram-Schmidt process on these vectors:

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - (-2)\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \mathbf{x}_3 - \frac{3}{2}\mathbf{v}_1 - \left(-\frac{1}{2}\right)\mathbf{v}_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$$

$$\text{Thus an orthogonal basis for } W \text{ is } \left\{ \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

10. Call the columns of the matrix $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ and perform the Gram-Schmidt process on these vectors:

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - (-3)\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \mathbf{x}_3 - \frac{1}{2}\mathbf{v}_1 - \frac{5}{2}\mathbf{v}_2 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}$$

$$\text{Thus an orthogonal basis for } W \text{ is } \left\{ \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix} \right\}.$$

11. Call the columns of the matrix $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ and perform the Gram-Schmidt process on these vectors:

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - (-1)\mathbf{v}_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \mathbf{x}_3 - 4\mathbf{v}_1 - \left(-\frac{1}{3}\right)\mathbf{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}$$

$$\text{Thus an orthogonal basis for } W \text{ is } \left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix} \right\}.$$

12. Call the columns of the matrix $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ and perform the Gram-Schmidt process on these vectors:

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{x}_2 - 4\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \mathbf{x}_3 - \frac{7}{2}\mathbf{v}_1 - \frac{3}{2}\mathbf{v}_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ -3 \end{bmatrix}$$

$$\text{Thus an orthogonal basis for } W \text{ is } \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ -3 \end{bmatrix} \right\}.$$

13. Since A and Q are given,

$$R = Q^T A = \begin{bmatrix} 5/6 & 1/6 & -3/6 & 1/6 \\ -1/6 & 5/6 & 1/6 & 3/6 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}$$

14. Since A and Q are given,

$$R = Q^T A = \begin{bmatrix} -2/7 & 5/7 & 2/7 & 4/7 \\ 5/7 & 2/7 & -4/7 & 2/7 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{bmatrix} = \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix}$$

Question 6:

1. To find the normal equations and to find $\hat{\mathbf{x}}$, compute

$$A^T A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

a. The normal equations are $(A^T A)\mathbf{x} = A^T \mathbf{b}$: $\begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$.

b. Compute

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}^{-1} \begin{bmatrix} -4 \\ 11 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 22 & 11 \\ 11 & 6 \end{bmatrix} \begin{bmatrix} -4 \\ 11 \end{bmatrix} \\ &= \frac{1}{11} \begin{bmatrix} 33 \\ 22 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{aligned}$$

2. To find the normal equations and to find $\hat{\mathbf{x}}$, compute

$$A^T A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} -24 \\ -2 \end{bmatrix}$$

a. The normal equations are $(A^T A)\mathbf{x} = A^T \mathbf{b}$: $\begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -24 \\ -2 \end{bmatrix}$.

b. Compute

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}^{-1} \begin{bmatrix} -24 \\ -2 \end{bmatrix} = \frac{1}{56} \begin{bmatrix} 10 & -8 \\ -8 & 12 \end{bmatrix} \begin{bmatrix} -24 \\ -2 \end{bmatrix} \\ &= \frac{1}{56} \begin{bmatrix} 224 \\ 168 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \end{aligned}$$

3. To find the normal equations and to find $\hat{\mathbf{x}}$, compute

$$A^T A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$$

a. The normal equations are $(A^T A)\mathbf{x} = A^T \mathbf{b}$: $\begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$

b. Compute

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ -6 \end{bmatrix} = \frac{1}{216} \begin{bmatrix} 42 & -6 \\ -6 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ -6 \end{bmatrix} \\ &= \frac{1}{216} \begin{bmatrix} 288 \\ -72 \end{bmatrix} = \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix} \end{aligned}$$

4. To find the normal equations and to find $\hat{\mathbf{x}}$, compute

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix}$$

a. The normal equations are $(A^T A)\mathbf{x} = A^T \mathbf{b}$: $\begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix}$

b. Compute

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 14 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 11 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 14 \end{bmatrix} \\ &= \frac{1}{24} \begin{bmatrix} 24 \\ 24 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{aligned}$$

5. To find the least squares solutions to $A\mathbf{x} = \mathbf{b}$, compute and row reduce the augmented matrix for the system $A^T A\mathbf{x} = A^T \mathbf{b}$:

$$\begin{bmatrix} A^T A & A^T \mathbf{b} \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 & 14 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

so all vectors of the form $\hat{\mathbf{x}} = \begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ are the least-squares solutions of $A\mathbf{x} = \mathbf{b}$.

6. To find the least squares solutions to $A\mathbf{x} = \mathbf{b}$, compute and row reduce the augmented matrix for the system $A^T A\mathbf{x} = A^T \mathbf{b}$:

$$\begin{bmatrix} A^T A & A^T \mathbf{b} \end{bmatrix} = \begin{bmatrix} 6 & 3 & 3 & 27 \\ 3 & 3 & 0 & 12 \\ 3 & 0 & 3 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

so all vectors of the form $\hat{\mathbf{x}} = \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ are the least-squares solutions of $A\mathbf{x} = \mathbf{b}$.

7. From Exercise 3, $A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}$, and $\hat{\mathbf{x}} = \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix}$. Since

$$A\hat{\mathbf{x}} - \mathbf{b} = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ 3 \\ -1 \end{bmatrix}$$

the least squares error is $\|A\hat{\mathbf{x}} - \mathbf{b}\| = \sqrt{20} = 2\sqrt{5}$.

8. From Exercise 4, $A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$, and $\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Since

$$A\hat{\mathbf{x}} - \mathbf{b} = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

the least squares error is $\|A\hat{\mathbf{x}} - \mathbf{b}\| = \sqrt{6}$.

9. (a) Because the columns $\mathbf{a}_1$ and $\mathbf{a}_2$ of A are orthogonal, the method of Example 4 may be used to find $\hat{\mathbf{b}}$, the orthogonal projection of $\mathbf{b}$ onto Col A :

$$\hat{\mathbf{b}} = \frac{\mathbf{b} \cdot \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_1} \mathbf{a}_1 + \frac{\mathbf{b} \cdot \mathbf{a}_2}{\mathbf{a}_2 \cdot \mathbf{a}_2} \mathbf{a}_2 = \frac{2}{7} \mathbf{a}_1 + \frac{1}{7} \mathbf{a}_2 = \frac{2}{7} \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} + \frac{1}{7} \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

(b) The vector $\hat{\mathbf{x}}$ contains the weights which must be placed on $\mathbf{a}_1$ and $\mathbf{a}_2$ to produce $\hat{\mathbf{b}}$. These weights are easily read from the above equation, so $\hat{\mathbf{x}} = \begin{bmatrix} 2/7 \\ 1/7 \end{bmatrix}$.

15. The least squares solution satisfies $R\hat{\mathbf{x}} = Q^T \mathbf{b}$. Since $R = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}$ and $Q^T \mathbf{b} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$, the augmented matrix for the system may be row reduced to find

$$\begin{bmatrix} R & Q^T \mathbf{b} \end{bmatrix} = \begin{bmatrix} 3 & 5 & 7 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -1 \end{bmatrix}$$

and so $\hat{\mathbf{x}} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$ is the least squares solution of $A\mathbf{x} = \mathbf{b}$.

16. The least squares solution satisfies $R\hat{\mathbf{x}} = Q^T \mathbf{b}$. Since $R = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}$ and $Q^T \mathbf{b} = \begin{bmatrix} 17/2 \\ 9/2 \end{bmatrix}$, the augmented matrix for the system may be row reduced to find

$$\begin{bmatrix} R & Q^T \mathbf{b} \end{bmatrix} = \begin{bmatrix} 2 & 3 & 17/2 \\ 0 & 5 & 9/2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2.9 \\ 0 & 1 & .9 \end{bmatrix}$$

and so $\hat{\mathbf{x}} = \begin{bmatrix} 2.9 \\ .9 \end{bmatrix}$ is the least squares solution of $A\mathbf{x} = \mathbf{b}$.